Sharp subsidies for truthful production and network cooperation

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Abstract

Owners of complementary productive resources may gain by withholding capacity when a sharing rule selects efficient core allocations. We ask how much external subsidy suffices to retain every coalition's production guarantee while making each fixed owner's payment nondecreasing in its reported resource bundle. For any technology represented by the minimum of K fixed nonnegative linear forms, we construct a deterministic rule satisfying both requirements at all nonnegative real report profiles. Its total payment is at most $(1 + o(1))T \log K / \log \log K$, where T is grand-coalition production. A matching lower bound holds for the worst case with at most K structural cuts even on simple directed acyclic flow networks with distinct strategic edge owners. Thus the leading constant is one. The upper combines capped supporting prices selected without the recipient's report with endpoint marginal payments. The lower transports payments across unilateral report changes and explicitly accounts for auxiliary owners. The incentive model permits binding componentwise underreports and payment-only utility. The bound is uniform in owner count and resource dimension, but K may be exponential in a compact description. Exact finite evaluation exists; polynomial preprocessing is not asserted. The result isolates the cost of linking coalition-covering allocations across reports in games whose individual profiles already admit efficient core allocations. For every finite monotone concave homogeneous technology we also obtain total-payment factors $n\sqrt{2/(n+1)}$ and $1 + 2\sqrt{d}$ in fixed owner count n and resource dimension d. Their unrestricted optimal rates remain open.

Keywords: production game; mechanism design; capacity reporting; coalitional stability; subsidy; flow game.
JEL classification: C71; D82; D47.

1 Introduction

Resource owners can jointly create value that none can produce alone. A sharing rule must then address two distinct problems: retaining the coalition and eliciting the resources available to it. At a given capacity profile, competitive supporting prices can allocate the entire production value while meeting every coalition's outside option. The difficulty is choosing these allocations consistently across profiles. An owner who receives less after revealing additional resources has an incentive to withhold them.

We measure the external budget needed to reconcile these requirements. Owners, resource types and technology are fixed. Each owner reports a nonnegative bundle and can commit no more than its true resources. Its utility is its payment; resource use has no private cost. A single deterministic rule must work at every nonnegative real profile, including zeros and ties. Payments are nonnegative, cover every coalition's production value, and are nondecreasing in the reporting owner's entire

bundle with the other reports held fixed. These assumptions describe capacity disclosure by fixed partners; they do not model hidden opportunity costs or the choice of who joins.

The production function is a minimum of K fixed nonnegative linear forms. This includes finite nonnegative linear production programs and single-commodity flow networks. A form can represent supporting resource prices; in a network it is a structural cut. Grand production is denoted by T. If total payment is at most BT, the external subsidy is at most $(B-1)T$. Coverage includes the grand coalition, so total payments cannot be below T at positive-value profiles.

1.1 Result and economic interpretation

We prove that the optimal worst-case total-payment factor is

$$(1+o(1))\frac{\log K}{\log\log K}$$

for technologies with at most K forms. The same leading constant is necessary even on simple directed acyclic networks with one strategic owner per edge. Theorem 2.1 provides an explicit finite-K bound and a mechanism. Theorem 4.1 supplies the matching asymptotic lower bound for the worst case with at most K cuts. Owner and resource counts are unrestricted finite integers in both statements.

The lower bound means that no universal constant subsidy factor preserves all coalition guarantees and truthful capacity disclosure throughout this class. This is not caused by an empty core at a fixed profile. For any minimizing form w, the efficient payments $w \cdot b_i$ cover every coalition. Subsidy is required because payments must also satisfy incentive inequalities between profiles. Our lower bound connects those inequalities using actual unilateral changes by the original owners. All auxiliary edges in its network construction belong to strategic owners and can have arbitrary reports; no free infrastructure is introduced.

The upper bound selects supporting prices after randomly deleting whole owners. A recipient's excluded price does not depend on its own report, and its contribution is capped at grand production. Endpoint marginal payments repair coalition coverage. Averaging over a short interval of deletion probabilities controls the additional payment. Randomness defines a finite expectation: the implemented rule is deterministic. The proof uses a union bound over the fixed forms, which explains the technology-complexity parameter.

A small K gives a budget guarantee independent of the number of owners. However, K can be exponential in an encoded network or production program. Theorem 3.1 gives separate bounds of order $\sqrt{n}$ and $\sqrt{d}$, including technologies without a finite-form representation. These bounds improve the project's earlier owner-count and dimension estimates, but their unrestricted optimal rates remain open. No polynomial-time preprocessing algorithm is asserted. It also makes no incentive claim for resource-use costs, overreports, coordinated deviations, false names, or changes in the owner population. These are separate economic questions.

1.2 Related work and contribution boundary

Dual-price core allocations are classical in linear production games (Owen, 1975; Samet and Zemel, 1984). The present contribution concerns selecting payments across reports under a uniform budget bound. Fixed-game cost of stability (Bachrach et al., 2018) studies subsidy for coalitional stability; here efficient cores already exist profile by profile.

The closest strategic comparison is Huang, Mei, Yang and Cao's *Mechanism Design in Max-Flows* (2025, inspected arXiv:2502.08248v1, Sections 2 and 4). It uses binding capacity underreports and

payment utility, establishes an obstruction to truthful efficient core selection, and studies efficient truthful mechanisms without universal core membership, including its Shapley rule (Proposition 1). We retain every coalition inequality and permit subsidy. We do not claim their split/merge or other-owner monotonicity guarantees.

Immorlica et al. (2008) impose cross-monotonicity when other owners join. Brânzei et al. (2007) define type-monotonic schemes on participating subgames of a unit-resource market (Definition 1 and Theorems 5–6 of the final 2007 text). Those comparisons change the population; ours changes one fixed owner's reported bundle. No equivalence via strategic cloning is used.

Amigo et al. (2016) do use own-resource monotonicity for a fixed alliance. Their inspected author version defines a strictly concave network-utility model and studies projection onto the full efficient core. Its Theorem 1 proves own-resource monotonicity of the Shapley value, while noting that this value can fail core membership. Its projection monotonicity evidence is computational, not a universal proof. Online Resource 1 supplies a strictly concave counterexample to a universal extension of that evidence, with a version-specific comparison. This auxiliary example is not an instance of our homogeneous main theorem.

Weighted multi-glove games and their total balancedness are established (Moretti and Norde, 2021). Their sharper special-case analysis, the connection to the continuous Banzhaf functional, and additional mechanism comparisons appear in Online Resource 1. Supporting prices, marginal contributions, concentration bounds and weighted counting are established tools. The proposed contribution is their combination into the joint all-profile coverage, incentive and sharp budget guarantee, with a matching lower bound under fixed ownership. The bounded source audit is not a certification of historical priority.

1.3 Organization

Section 2 gives the model, mechanism and complete upper proof. Section 3 gives complete owner-count and dimension proofs for general concave homogeneous production. Section 4 proves the complexity lower bound and explains why naive subdivision would destroy it. The conclusion separates the unrestricted parameter gaps from the restricted mechanism barriers and computational questions. Online Resource 1 is an indexed supplement containing the earlier hierarchy, general extensions and exact obstructions; its repeated foundational proofs make it independently readable. Online Resource 2 contains the scoped verification code and logs.

2 General production: a sharp complexity bound

There are n fixed owners and d fixed resource types. Owner i reports $b_i \in \mathbb{R}^d_+$; write $c = \sum_i b_i$. Fix a nonempty finite family $V = \{w^1, \dots, w^K\} \subseteq \mathbb{R}^d_+$, with a report-independent priority, and define

$$f(c) = \min_{w \in V} w \cdot c, \qquad v_b(S) = f\left(\sum_{i \in S} b_i\right), \qquad T = f(c).$$

A rule is eligible if its payments are nonnegative, cover every coalition, and are componentwise nondecreasing in each owner's whole bundle, with all other reports fixed. These conditions hold on all of $(\mathbb{R}^d_+)^n$. They express truthfulness for binding underreports and payment-only utility; there are no operating costs, overreport, false-name, population-change or coordinated-report guarantees. A factor C bounds *total payment* by CT; external subsidy is at most $(C-1)T$. The family V remains fixed across reports. Its size may be exponential in a compact description. All logarithms are natural.

Theorem 2.1 (General sharp-order upper bound)**.** *For every integer $j \geq 2$, there exists an eligible deterministic rule with factor*

$$C_j(K) = j + \frac{\log K + 2\log j + 1}{\alpha_j}, \qquad \alpha_j = \log \frac{j^3}{j^2 + j - 1} > 0. \tag{1}$$

With $L = \log K$ and $j = \max\{2, \lceil L/[\log(2+L)]^2 \rceil\}$, this factor is $(1 + o(1)) \log K / \log\log K$, uniformly over owner counts, dimensions, technologies and reports. Zero-value profiles receive zero; for $K = 1$ the constructed rule is efficient.

The mechanism. For $q \in (0, 1)$ let independent owner bits satisfy $\mathbb{P}(B_i = 0) = q$. Choose the first minimizer of the *uncapped* objective,

$$W(b, B) = \operatorname*{first\,argmin}_{w \in V} \sum_i B_i (w \cdot b_i), \qquad a_i(w, b) = \min\{w \cdot b_i, T\}.$$

Only the payment contribution a_i is capped. Put

$$z_i^q = \mathbb{E} a_i(W, b), \qquad p_i^q = \mathbb{E}[a_i(W, b) \mid B_i = 0], \tag{2}$$

$$m_i^q = \mathbb{E}_{B_{-i}} \left[f\left(b_i + \sum_{h \neq i} B_h b_h \right) - f\left(\sum_{h \neq i} B_h b_h \right) \right], \qquad r_i^q = q p_i^q + (1 - q) m_i^q. \tag{3}$$

The final payment is the deterministic finite expectation

$$M_i(b) = j \int_0^{1/j} r_i^q(b)\, dq. \tag{4}$$

These quantities are finite polynomials in q at a fixed profile; no continuity or generic-position assumption about the reports is required.

Proof of Theorem 2.1. Conditional on $B_i = 0$, every score and the selected form are independent of b_i. For that fixed nonnegative form, both $w \cdot b_i$ and T are nondecreasing in b_i, so their minimum is nondecreasing. In m_i^q the off term is independent of b_i and the on term is nondecreasing. The nonnegative fixed mixtures preserve this property and nonnegativity.

For any w, S, if some $i \in S$ has $w \cdot b_i \geq T$, then $\sum_{i \in S} a_i(w, b) \geq T \geq v_b(S)$. Otherwise no term in S is capped, and the sum is $w \cdot \sum_{i \in S} b_i \geq v_b(S)$. Thus every capped form covers coalitions. Fix B_{-i} and let w_1 be the on minimizer. The off optimum is at most the off objective of this same form, whence

$$f\left(b_i + \sum_{h \neq i} B_h b_h \right) - f\left(\sum_{h \neq i} B_h b_h \right) \geq w_1 \cdot b_i = a_i(w_1, b). \tag{5}$$

The equality holds because the contribution is a nonnegative summand of the selected on value, which is at most T. Averaging proves $r_i^q \geq z_i^q$, and hence coverage for r^q and M.

For the budget define $g(q) = \mathbb{E} f(\sum_i B_i b_i)$, $Z(q) = \sum_i z_i^q$, and $R(q) = \sum_i r_i^q$. Capping leaves every open contribution unchanged:

$$\sum_i B_i a_i(W, b) = f\left(\sum_i B_i b_i \right).$$

Consequently $q\sum_i p_i^q = Z - g$. Differentiating the finite mask polynomial gives $\sum_i m_i^q = -g'$, so $R = Z - g - (1-q)g'$. Since $g(0) = T$ and $g \geq 0$, for $0 < \eta < 1$ integration by parts gives the exact identity

$$\int_0^\eta R\,dq = \int_0^\eta Z\,dq + T - (1-\eta)g(\eta) - 2\int_0^\eta g\,dq \leq \int_0^\eta Z\,dq + T. \tag{6}$$

Both the sign and the factor two in this identity matter.

Suppose $T > 0$. For each fixed form let $d_i(w) = a_i(w,b)/T \in [0,1]$ and $\rho_w = \sum_i d_i(w)$. The selected form satisfies $\sum_i B_i d_i(W) \leq 1$. For $\lambda > 0$ put $\alpha = -\log(q + (1-q)e^{-\lambda})$. The convex function $u \mapsto \log(q + (1-q)e^{-\lambda u})$ lies below its endpoint chord $-\alpha u$ on $[0,1]$. Independence and Markov's inequality give

$$\mathbb{P}\left(\sum_i B_i d_i(w) \leq 1\right) \leq e^{\lambda}\prod_i (q + (1-q)e^{-\lambda d_i(w)}) \leq e^{\lambda - \alpha\rho_w}.$$

A union over the fixed family and integration of the resulting tail yield

$$\mathbb{P}(\rho_W \geq x) \leq \min\{1, Ke^{\lambda-\alpha x}\}, \qquad \frac{Z(q)}{T} \leq \frac{\log K + \lambda + 1}{\alpha}. \tag{7}$$

Take $\eta = 1/j$ and $\lambda = 2\log j$. For $q \leq 1/j$, $q + (1-q)e^{-\lambda} \leq (j^2 + j - 1)/j^3$, so (6)–(7) give (1). For the displayed choice of j, as $L \to \infty$, $j \sim L/(\log L)^2$, $\alpha_j \sim \log L$, and $j = o(L/\log L)$, proving the asserted asymptotic.

If $T = 0$, all masked production values and capped contributions vanish. If $K = 1$, no contribution is capped, $p_i^q = m_i^q = w \cdot b_i$, and $M_i = w \cdot b_i$. Finally each finite-mask term is $q^{n-1-|S|}(1-q)^{|S|}[qa_i(W(b,1_S),b) + (1-q)(v_b(S\cup\{i\}) - v_b(S))]$. Its integral is a finite sum with rational coefficients when j is integer. This gives exact rational payments for rational input, without an efficiency claim about enumeration of masks or forms. □

Corollary 2.2 (Production LPs and networks)**.** *The theorem applies to $f(c) = \max\{\pi \cdot x : x \geq 0, Ax \leq c\}$ for fixed $A \geq 0$, with every activity column nonzero and fixed positive profits π. Here V can be the vertices of $\{w \geq 0 : A^T w \geq \pi\}$. For single-commodity directed flow, one may instead use the characteristic vectors of the inclusion-minimal structural source–sink edge cuts.*

Proof. Each activity is bounded by a positive entry in its column, so every feasible production set is compact at finite c. The dual is nonempty, pointed and polyhedral. Strong duality and attainment at a vertex give the claimed finite min-linear representation, including boundary capacities. Network max-flow/ min-cut gives the cut representation directly. Redundant nonminimal cuts may be removed once and for all because capacities are nonnegative. All owners and forms remain fixed across profiles. A positive-profit zero-resource activity is excluded because it would make production unbounded. □

Dual core allocations are established ingredients (Owen, 1975; Samet and Zemel, 1984). General dual vertices may be fractional; their count is not an ordinary network-cut count. Constraints introducing capacity-independent dual offsets are outside the homogeneous model here.

Proposition 2.3 (Why the payment cap is essential)**.** *Removing the cap from the own-excluded contribution in* (2) *makes the uniform-q hybrid unbounded even with two owners and two supporting forms.*

Proof. Take unit-profit activity columns $(1,1)$ and $(M,0)$, $M>1$, and two fixed owners with bundles $b_1=(M,0)$ and $b_2=(0,1)$, so $c=(M,1)$. The two dual vertices are $(1,0)$ and $(1/M,1-1/M)$, and $T=2-1/M$. If owner 1 is off and owner 2 on, $(1,0)$ uniquely minimizes the raw score, giving uncapped contribution M. Thus $r_1^q \geq q(1-q)M$; with $j=2$ its average is at least $M/6$, an unbounded ratio to T. This obstructs only the uncapped rule. On the full two-owner bundle domain, the Shapley payments $P_i(b)=[f(b_i)+T-f(b_{-i})]/2$ are efficient and own-monotone. They are nonnegative by monotonicity of f, and cover singleton coalitions because $f(b_1+b_2)\geq f(b_1)+f(b_2)$. This last inequality follows by minimizing nonnegative linear forms. Thus an efficient eligible alternative exists without restricting each owner's report to a single resource. □

3 Bounds in owner count and resource dimension

The number of supporting forms can be much larger than the economic description. We now obtain bounds directly in the numbers of fixed owners and resources. Here the production technology may be any finite-valued monotone concave function $f:\mathbb{R}^d_+\to\mathbb{R}_+$ homogeneous of degree one; a finite min-linear representation is not required. The report space and eligibility conditions remain unchanged. In particular, there is no additional fairness floor.

Theorem 3.1 (General owner and dimension bounds)**.** *For every fixed $n,d\geq 1$ and every such technology, there exists an eligible deterministic rule on all nonnegative real report profiles with*

$$\sum_i M_i(b) \leq \min\left\{d,\ n\sqrt{\frac{2}{n+1}},\ 1+2\sqrt{d}\right\}T. \tag{8}$$

The individual bounds are uniform in the other parameter. They also hold when f has downward discontinuities at the boundary. More precisely, for every fixed $0<a<1$ the supporting-price/endpoint rule averaged over $q\in[0,a]$ has both bounds

$$B_n(a)=\frac{na}{2}+\frac{n[1-(1-a)^{n+1}]}{(n+1)a}, \tag{9}$$

$$D_d(a)=\frac{1+2d[-a-\log(1-a)]}{a}. \tag{10}$$

The owner expression remains valid at $a=1$ by polynomial integration. No polynomial-time evaluation or preprocessing claim is made.

3.1 Supporting prices on boundary faces

For $x\neq 0$, restrict f to its coordinate face $F_x=\{z\geq 0 : z_e=0 \text{ whenever } x_e=0\}$. At the relative-interior point x, finite concavity gives a supergradient u. Monotonicity forces each active coordinate of u to be nonnegative, and homogeneity applied above and below x gives $u\cdot x=f(x)$. For a boundary point $z\in F_x$, apply the supporting inequality at $(1-t)x+tz$, $0<t<1$, and the concavity lower bound on that segment to obtain $u\cdot z\geq f(z)$. Thus boundary continuity is unnecessary. Choose the unique minimum-norm vector in the nonempty closed convex set of such nonnegative supports on F_x. Extend its linear form to

$$\ell_x(z)=\begin{cases} u\cdot z, & z\in F_x,\\ +\infty, & z\notin F_x.\end{cases}$$

Set $\ell_0(0) = 0$ and $\ell_0(z) = +\infty$ for $z \neq 0$. This is one fixed selection depending only on x. For finite min-linear technologies the original fixed-priority global minimizing form is an alternative; the bounds below apply to either choice.

For any fixed mask $B \subseteq N$, put $x_B = \sum_{i \in B} b_i$ and $A_i(B) = \min\{\ell_{x_B}(b_i), T\}$. This vector covers every coalition: if any member's raw price reaches T, its capped payment alone suffices; otherwise all relevant prices are finite on the support face and their sum majorizes the coalition's worth. For $i \in B$,

$$f(x_B) - f(x_B - b_i) \geq \ell_{x_B}(b_i),$$

because the support touches at x_B and majorizes $f(x_B - b_i)$. The marginal is at most T. Replace open owners' prices by these marginals and keep $A_i(B)$ for closed owners. Coverage is preserved. A closed owner's selected price excludes its own report, while its raw price and cap are nondecreasing in that report. An open owner's marginal has a fixed off term and a nondecreasing on term. This proves eligibility for every fixed mask and every report-independent mixture of masks.

With independent missing probability q, retain the notation p_i^q, m_i^q, r_i^q from (2)–(3), now using ℓ_{x_B}. At a fixed profile every capped mask payment is finite, so the expectations are finite polynomials in q. Define $M_i^a = a^{-1} \int_0^a r_i^q \, dq$. The endpoints have measure zero, including when $a = 1$. Writing $g(q) = \mathbb{E} f(x_B)$ and $C(q) = q \sum_i p_i^q$, differentiation of the finite mask polynomial gives

$$\sum_i r_i^q = C(q) - (1-q)g'(q), \qquad \int_0^a -(1-q)g'(q)\, dq = T - (1-a)g(a) - \int_0^a g(q)\, dq. \tag{11}$$

No derivative of f with respect to reports occurs here.

3.2 Owner-count bound

Every closed payment is at most T, whence $C(q) \leq nqT$. The all-open mask alone gives $g(q) \geq (1-q)^n T$. Substituting both estimates in (11), integrating and dividing by a yields

$$\sum_i M_i^a \leq \left(\frac{na}{2} + \frac{n[1 - (1-a)^{n+1}]}{(n+1)a} \right) T.$$

Dropping the bracketed power term gives $B_n(a) \leq na/2 + n/[(n+1)a]$. The fixed choice $a = \sqrt{2/(n+1)} \leq 1$ equalizes these terms and proves the owner bound in (8). For $n = 1$ the exact expression is $B_1(a) = 1$. The former $O(n/\log n)$ upper bound recorded in the supplement is therefore superseded by $(\sqrt{2} + o(1))\sqrt{n}$.

3.3 Resource-dimension bound

The following scalar identity controls contributions from closed owners without counting owners or supporting forms.

Lemma 3.2 (Bernoulli reciprocal bound)**.** *Let $u_i \geq 0$, independently retain each i with probability $p = 1 - q$, and write $S_B = \sum_{i \in B} u_i$. Define $\min\{u_i/S_B, 1\}$ to be zero when $u_i = 0$ and one when $u_i > 0 = S_B$. Then*

$$\mathbb{E} \sum_{i \notin B} \min\{u_i/S_B, 1\} \leq \frac{2q}{1-q}.$$

Proof. With zero-over-zero interpreted as zero, $\min\{u/S, 1\} \leq 2u/(u+S)$. Conditioning on the own bit gives

$$\mathbb{E} \sum_{i \notin B} \frac{u_i}{u_i + S_B} = \frac{q}{p} \mathbb{E} \sum_{i \in B} \frac{u_i}{S_B} = \frac{q}{p} \mathbb{P}(S_B > 0) \leq \frac{q}{p}.$$

In the first equality both conditioned ratios are $u_i/(u_i + \sum_{j\neq i} B_j u_j)$; the other bits have the same law. □

Suppose $T > 0$. For every mask and owner,

$$\frac{A_i(B)}{T} \leq \sum_{e=1}^{d} \min\left\{\frac{b_{ie}}{(x_B)_e}, 1\right\}, \tag{12}$$

with the lemma's conventions. If b_i has a positive coordinate outside F_{x_B}, the right side is at least one. Otherwise the selected nonnegative support satisfies $\sum_e u_e(x_B)_e = f(x_B) \leq T$, so each active term obeys $u_e b_{ie}/T \leq b_{ie}/(x_B)_e$. Apply $\min\{\sum_e z_e, 1\} \leq \sum_e \min\{z_e, 1\}$. The case $x_B = b_i = 0$ is immediate. The same argument applies to a finite global minimizing form, since off-face prices remain capped by T.

Sum (12) over closed owners and use Lemma 3.2 separately in each resource coordinate: $C(q) \leq 2dqT/(1-q)$. The endpoint integral in (11) is at most T. Since $\int_0^a q/(1-q)\,dq = -a - \log(1-a)$, this proves (10). Also

$$-a - \log(1-a) = \sum_{k\geq 2} \frac{a^k}{k} \leq \frac{a^2}{2(1-a)}, \qquad D_d(a) \leq \frac{1}{a} + \frac{da}{1-a}.$$

Choosing $a = 1/(1+\sqrt{d})$ gives $1 + 2\sqrt{d}$. At $T = 0$, monotonicity makes all coalition values, caps and endpoint marginals vanish, completing these cases.

3.4 Baseline, selection and remaining gap

The grand marginal $U_i = f(c) - f(c - b_i)$ is eligible: it is own-monotone and dominates the support allocation at c. Put $\alpha_i = \max_{e:c_e>0} b_{ie}/c_e$ when $T > 0$. Then $c - b_i \geq (1-\alpha_i)c$, so $U_i \leq \alpha_i T$ and $\sum_i \alpha_i \leq d$. This proves the dT baseline. Choosing an entire rule using the fixed public n, d before reports proves (8); no pointwise minimum of payment vectors is used. For a finite K representation, the complexity bound in Theorem 2.1 is another available choice.

The scalar leading constant two is already sharp as $q \downarrow 0$: two equal positive weights give expected closed ratio exactly $2q$. This does not prove optimality of the dimension constant for the full mechanism. The unrestricted owner and dimension rates remain open between the earlier logarithmic-over-logarithmic lower bounds and the square-root uppers above. The lower bounds and their parameter translations are proved in Online Resource 1, theorems titled "Owner-count range" and "Concave dimension gap". Online Resource 1 gives matching square-root growth obstructions only for specified fixed-mask, common-support and endpoint architectures, including a maximum-coalition-marginal cap variant. Those restrictions are essential; they are not lower bounds for arbitrary eligible payment rules.

Square-root payment bounds also arise in static cost-of-stability theory: Bachrach et al. (2009, arXiv:0907.4385v1, Theorem 3.5) and the published Bachrach et al. (2018, Theorem 7) give a total-funding factor $\sqrt{n}$ for superadditive games without the cross-report incentive requirement. Every profile in our model already admits an efficient support allocation, so its static cost of stability is zero. The new bounds concern one rule linking those profiles.

4 A matching lower bound on simple networks

A network has one distinct fixed strategic owner per edge, and coalition worth is its maximum source–sink flow using only its owned edges. Structural cuts are inclusion-minimal edge sets

disconnecting source from sink; their number depends on the fixed graph, not its reported capacities. Define $\mathcal{B}_{\mathrm{dag}}(K)$ as the supremum, over simple finite DAGs having at most K structural cuts, of the best all-profile eligible total-payment factor on that graph. The analogous quantity for finite nonnegative min-linear technologies with at most K forms is $\mathcal{B}_{\mathrm{prod}}(K)$; owner counts and dimensions are unrestricted finite integers.

Theorem 4.1 (Sharp first-order worst-case factors)**.** *As $K \to \infty$,*

$$\mathcal{B}_{\mathrm{dag}}(K) = (1+o(1))\frac{\log K}{\log\log K}, \qquad \mathcal{B}_{\mathrm{prod}}(K) = (1+o(1))\frac{\log K}{\log\log K}.$$

The lower holds along an explicit sequence of simple DAGs at their exact cut counts, and for the worst case with at most *K cuts. It does not assert an exact-count padding theorem for every integer K.*

Proof. The upper follows from Theorem 2.1. We give the simple-DAG lower, which also supplies the production lower. Fix integers $d_0 \geq 1$, $m > d_0$, $R \geq 1$, and $k \geq 2$. (Here d_0 is hierarchy height, not resource dimension.) One module with terminals a, b has vertices $L_1, \ldots, L_R$, $u_1, \ldots, u_m$, $v_1, \ldots, v_m$, $W_1, \ldots, W_R$ and edges

$$aL_s, \quad L_s u_j, \quad u_j v_j, \quad v_j W_t, \quad W_t b \quad (1 \leq s,t \leq R,\ 1 \leq j \leq m).$$

Take k copies in series by identifying consecutive terminals. Each middle edge $u_j v_j$ has report in $\{1, m\}$; all auxiliary edges have report $H = m^2$. Every edge, including every auxiliary edge, is a strategic original owner whose full report domain is nonnegative. Only the finite profiles used in this proof keep auxiliary reports fixed.

A type-h module has h high middle edges, $0 \leq h \leq d_0$. For any subset J of middle edges, choose one left fan and one right fan, and include the source/final fan edges plus the incident edges supporting exactly J. This actual coalition has module flow $\sum_{j \in J} c_j$: the middle cut bounds it above, and the selected branches carry it because their total is at most m^2. In particular the full graph has

$$T = m + (m-1) \min_{1 \leq l \leq k} H_l. \tag{13}$$

Choosing the two fans independently uniformly, any particular auxiliary edge is included with probability at most $1/R$.

Symmetrize an arbitrary eligible factor-B rule over module permutations and branch permutations, relabeling each branch's incident auxiliary edges. These are permutations of the fixed owner set preserving every coalition's flow at all profiles: module flows combine by their minimum, so even module permutations preserve coalition values. Averaging preserves all required inequalities. At a histogram $\nu = (\nu_0, \ldots, \nu_{d_0})$ of module types, write $x_h(\nu), z_h(\nu)$ for low and high variable-owner payments, and

$$Y(\nu) = \sum_h \nu_h[(m-h)x_h(\nu) + h z_h(\nu)], \qquad Q(\nu) = \text{total auxiliary-owner payment.}$$

No auxiliary-payment symmetry is assumed; $Y + Q \leq BT$.

First choose one low variable in each module, supported by random fans. Averaged coalition coverage and nonnegativity give $\sum_h \nu_h x_h + Q/R \geq 1$. Next put $q_0 = m - d_0 + 1$ and choose q_0 low variables in each type below d_0, and one high variable in each type d_0. These exist because $m > d_0$.

The resulting coalition has worth at least q_0, giving $q_0 \sum_{h<d_0} \nu_h x_h + \nu_{d_0} z_{d_0} + Q/R \geq q_0$. Multiply the first inequality by $m - d_0$ and the second by d_0. With $A = (d_0 + 1)m - d_0^2$,

$$A \leq A \sum_{h<d_0} \nu_h x_h + \nu_{d_0}[(m - d_0)x_{d_0} + d_0 z_{d_0}] + (m/R)Q. \tag{14}$$

Fix $\alpha > 0$, set $\lambda = A/\alpha$, $Z = \sum_{h=0}^{d_0} \lambda^h/h!$, and $p_h = \lambda^h/(h!Z)$. Average profiles with independent module types of probabilities p_h; let $\mu(\nu)$ be their multinomial histogram weights. Raising one actual low variable owner from 1 to m gives $x_h(\nu) \leq z_{h+1}(\nu')$ at $\nu' = \nu - e_h + e_{h+1}$. Every other report, including every auxiliary report, is unchanged. The exact identity

$$\mu(\nu)\nu_h = (p_h/p_{h+1})\mu(\nu')\nu'_{h+1}, \qquad Ap_h/p_{h+1} = \alpha(h + 1)$$

transports only this changing owner's payment. Averaging (14) therefore yields

$$\begin{aligned} A &\leq \alpha\mathbb{E} \sum_{h=1}^{d_0} h\nu_h z_h + \mathbb{E}\nu_{d_0}[(m - d_0)x_{d_0} + d_0 z_{d_0}] + (m/R)\mathbb{E}Q \\ &\leq (1 + \alpha)\mathbb{E}Y + (m/R)\mathbb{E}Q \leq \max\{1 + \alpha, m/R\}B\mathbb{E}T. \end{aligned} \tag{15}$$

The last step uses the disjoint budgets Y, Q; their coefficients are not added. By the tail-sum formula and $p_0 = 1/Z$,

$$\mathbb{E}T = m + (m - 1) \sum_{r=1}^{d_0} \left(\sum_{h=r}^{d_0} p_h \right)^k \leq m[1 + d_0 e^{-k/Z}]. \tag{16}$$

Let $C_{m,R}$ be the number of structural cuts of one module. Any minimal cut is the outgoing cut of its reachable source side after deletion, so $1 \leq C_{m,R} \leq 2^{2R+2m}$. A minimal cut in a series composition lies entirely in one module: otherwise each module retains a path, or an edge removed elsewhere is redundant. Hence the full cut count is exactly $K = kC_{m,R}$. For $d_0 \geq 2$ choose

$$g = \lceil \sqrt{\log(d_0 + 1)} \rceil, \quad m = d_0 g, \quad R = m, \quad \alpha = 1/g, \quad \ell = \lceil \log(d_0 + 1) \rceil, \quad k = \lceil 2\ell Z \rceil.$$

The separate multiplier ℓ, rather than g, ensures the tail error vanishes. Equations (15)–(16) give

$$B \geq \frac{d_0 + 1 - d_0/g}{(1 + 1/g)(1 + d_0 e^{-2\ell})} = (1 - o(1))d_0.$$

Here $\lambda = (d_0 + 1)d_0 g^2 - d_0^2 g > d_0$, so $\lambda^{d_0}/d_0! \leq Z \leq (d_0 + 1)\lambda^{d_0}/d_0!$. Stirling's estimate implies $\log k = (1 + o(1))d_0 \log d_0$; since $m = o(d_0 \log d_0)$,

$$\log k \leq \log K \leq \log k + 4m \log 2 = (1 + o(1))d_0 \log d_0.$$

This proves the lower along the sequence K_{d_0}. For any fixed $0 < \varepsilon < 1$ and large external cut budget K, take $d_0 = \lfloor (1 - \varepsilon) \log K / \log \log K \rfloor$. Then $K_{d_0} \leq K$ eventually, yielding lower coefficient $1 - \varepsilon$. Letting ε decrease to zero proves the at-most-K assertion. □

Proposition 4.2 (A subdivision that destroys the lower bound)**.** *Fully subdividing every edge of any finite directed network into two edges owned separately produces a network with an efficient eligible rule.*

Proof. For pair reports (a_e, b_e) set $d_e = \min(a_e, b_e)$. Choose the original network's fixed-priority minimum-cut core allocation $Q(d)$, which pays d_e on the selected cut and zero elsewhere. Pay $Q_e(d)$ to the higher reporting member of pair e, split equally at equality, and pay the lower member zero. Holding the mate's report fixed, the payment is zero below it, half the fixed Q_e at equality and the fixed Q_e above it: in the last two cases all virtual capacities are fixed. Thus it is own-monotone, even if Q is not. For a coalition S, let S^* contain the original edges whose two members both lie in S. Its value is exactly $v_d(S^*)$, and $p(S) \geq Q(S^*) \geq v_d(S^*)$. Total payment is T. This shows why the naive subdivision reduction fails and motivates the explicit auxiliary-budget accounting in the shared-fan construction. □

5 Implications, limitations and further questions

The result identifies technology complexity as one sharp measure of the budget required to combine truthful disclosure with coalition coverage. For every fixed finite representation, the upper holds simultaneously over all profiles and all owner bundles. The lower uses an increasing family of fixed technologies; it does not change the graph when an owner changes its report. Its asymptotic statement concerns a worst case with at most K cuts, not an exact-count construction for each integer K.

Two questions remain distinct from the sharp K theorem. First, Theorem 3.1 bounds the factor by $\min\{d, n\sqrt{2/(n+1)}, 1+2\sqrt{d}\}$, while the unrestricted lower bounds remain logarithmic over logarithmic in n and d. The supplement's matching square-root barriers apply to specified mask/support architectures; they do not close these unrestricted gaps. A better general rule must escape those restrictions. Second, rational finite technologies permit exact evaluation by finite enumeration, and the supplement develops finite sampling with all-profile certification. The certification preprocessing can be exponential. A polynomial preprocessing guarantee or a matching computational obstruction would be a separate contribution.

The main theorem deliberately isolates capacity disclosure. Private operating costs, endogenous participation, and coordinated reports require additional incentive constraints. A resource-sharing application should justify the payment-only assumption and its feasible deviations before interpreting the bound as a policy prescription.

Supplementary material. Online Resource 1 (`ESM_1.pdf`): indexed proofs of supplementary hierarchies, extensions and mechanism obstructions, including the original research lineage. Online Resource 2 (`ESM_2.zip`): scoped exact verification programs, execution logs, source hashes and reproduction instructions. Finite checks corroborate bounded identities and instances; the universal and asymptotic statements depend on the proofs above.

Statements and Declarations

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Data and code availability. No empirical dataset is used in the main theorem. The accompanying local reproducibility archive contains exact finite checks and their inputs, with version hashes and execution records. This candidate and archive have not been submitted or newly deposited by this preparation task.

AI assistance and responsibility. OpenAI Codex and Anthropic Claude assisted mathematical exploration, code, proof drafting and internal review across this research project. Internal model review is not human peer review or formal proof verification. The confirmed author(s) must independently assess the arguments, approve the text and references, confirm the declarations, and assume responsibility before any submission. Human specialist validation and historical priority remain open.

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